Type MS · Explain how diaphragm and intercostal muscles drive breathing

Answer 1 of 4 · Question 5 part a.iii · 6 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

5 (a) (i)		A = sternum; B = rib(cage); C = intercostal muscle;	ignore external/internal	3
	(ii)	• lungs; • diaphragm; • nose/trachea/bronchi/bronchus/bronchioles/alveoli;	in any order mark only first three	3
	(iii)	• diaphragm flattens; • ribs move up and out; • by contraction of intercostal muscles; • volume of thorax increases; • pressure in thorax/lungs decreases; • air forced/rushes/sucked/drawn, in (from outside);	reject moves down ignore external/internal ignore air flows in	6
(b)		• $C_6H_{12}O_6 + 6O_2$; • $6CO_2 + 6H_2O$; • energy/ATP;		3
				15 marks

Type MS · Explain how diaphragm and intercostal muscles drive breathing

Answer 1 of 4 · Question 5 part a.iii · 6 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how diaphragm and intercostal muscles drive breathing

Answer 2 of 4 · Question 8 part b.i · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

Number					
8	(a)	(i)	arrow pointing from top to bottom (1)		1
		(ii)	• provides energy/ATP (1) • for synthesis of acetylcholine/neurotransmitter (1) • movement of vesicles/exocytosis (1)		3
	(b)	(i)	• impulses can't pass to intercostal muscles/diaphragm (1) • unable to contract and move ribs/increase volume of thorax/chest (1)		2
		(ii)	• prevents breakdown of neurotransmitter/acetylcholine (1) • which can compete with curare for/binds to, receptor sites (1)		2

Total for this question = 8 marks

Type MS · Explain how diaphragm and intercostal muscles drive breathing

Answer 2 of 4 · Question 8 part b.i · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

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Type MS · Explain how diaphragm and intercostal muscles drive breathing

Answer 3 of 4 · Question 3 part c · 2 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

3 (a) (i)	C; (red blood cells)		1
(ii)	any three from • more carbon dioxide (1) • less oxygen (1) • moister (1) • warmer (1)		3
(iii)	• large surface area (1) • thin (walled)/wall one cell thick (1) • good blood supply (1) • moist lining (1)		3
(b)	• thin wall (1) • narrow lumen (1)	Allow references to only one blood cell passing through	2
(c)	• breathing (out)/exhalation more difficult/not smooth/shortness of breath/faster breathing rate (1) • as lungs would not squeeze out air/forced exhalation(1)		2

Total 11 marks

Type MS · Explain how diaphragm and intercostal muscles drive breathing

Answer 3 of 4 · Question 3 part c · 2 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how diaphragm and intercostal muscles drive breathing

Answer 4 of 4 · Question 7 part a · 4 marks

2024 JUN · Paper 01 · find part a in this mark scheme extract

7 (a)		• intercostal muscles contract; • ribs move up and out; • diaphragm contracts and flattens; • air is drawn into the lungs;	ignore ref to internal/external	(4)
(b)		• detect pH of blood/levels of CO_2; • to maintain homeostasis/regulate breathing/heart rate;		(2)
(c)	(i)	tidal (volume);		(1)
	(ii)	residual (volume);		(1)
	(iii)	vital capacity;		(1)

Total for Question 7 = 9 Marks